Joint Time-Frequency Resource Scheduling over CQF-based TSN-5G System

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2023 15th International Conference on Communication Software and Networks (ICCSN) | 979-8-3503-2936-0/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICCSN57992.2023.10297405

***Abstract*—Deterministic communication across the integrated time-sensitive networking (TSN) and 5G system (5GS) relies on the queue model and network resource scheduling. In this paper, we investigate the joint time-frequency resource scheduling problem in CQF-based TSN-5GS aiming to improve the number of schedulable flows and system load balance. To support the integration of CQF-based TSN and 5GS, we propose a time-aware packet forwarding strategy for 5GS. By managing the flow transmission time intervals and frequency resources, 5GS can seamlessly communicate with nearby CQF enabled TSN bridges. Given the complexity of network resource scheduling across the TSN-5GS, it is difficult for traditional algorithms to obtain the optimal scheduling results. Hence, we design a DRL-assisted time-frequency resource scheduling algorithm, namely DTF, to jointly allocate the time interval and frequency resources in the unit of flow. DTF embeds with a two-dimensional matrix of link capacity utilization to efficiently learn the relationship between different time intervals and capacity utilization. Experimental results show that our proposed DTF algorithm can increase the schedulable flow number by 54.2% and 3.5x than DeepCQF and Naive, respectively. DTF can also achieve a good trade-off between runtime and schedulability.**

***Index Terms*—CQF-based TSN-5GS, deep reinforcement learning, resource scheduling.**

I. Introduction

Recent years have witnessed a significant development of deterministic communication technologies. As an emerging paradigm for the deterministic communication, time-sensitive networking (TSN) over Ethernet is proposed by IEEE 802.1 TSN task group aiming to provide the time-sensitive applications with bounded latency, jitter, and high reliability. Several TSN queue models are defined by TSN standards, such as time-aware shaper (TAS), asynchronous traffic shaper (ATS), and cyclic queuing and forwarding (CQF) model for traffic shaping [1], [2]. Among these queue models, the CQF model has been widely adopted due to its ease of use features.

Although the CQF-based TSN can satisfy the deterministic communication demands of time-sensitive applications, it is still restricted into the wired network domain. With the rising of 5th generation (5G) wireless communication technology, more emerging 5G use cases, such as autonomous driving, remote surgery, virtual reality, etc., need a deterministic end-to-end communication guarantee [3]. Unfortunately, the 5G system (5GS) cannot support the determinism due to the inherent best-effort flow transmission mechanism. Hence, it is necessary to integrate CQF-based TSN with 5GS to extend the deterministic communication capability from wired network into wireless domain.

The 3rd Generation Partnership Project (3GPP) defines the architecture of integrated TSN-5GS, where 5GS is configured as a logical TSN bridge to seamlessly communicate with nearby TSN bridges [4]. Because the deterministic communication relies on network resource scheduling, many related works research the resource scheduling problem under the TSN-5GS architecture. Based on TAS mechanism, Ginthör *et al.* [5] proposed a window-based resource allocation method to improve the schedulability in TSN-5GS. Satka *et al.* designed a QoS-aware mapping algorithm to efficiently exchange flows between TSN and 5G network domain. Prados-Garzon *et al.* [6] presented a management framework to enable the TSN network slicing and inter-working with 5G backhaul. However, none of the related works considered resource scheduling in CQF-based TSN-5GS as the granularity of scheduling in CQF is more fine-grained and complex. Inspired by this, we investigate the joint time-frequency resource scheduling problem in CQF-based TSN-5GS aiming to improve the number of scheduled flows and system load balance.

Different from other queue models for TSN-5GS, CQF schedules the network resources at the granularity of each time interval within a hyper cycle. To support the integration of CQF-based TSN and 5GS, we propose a time-aware packet forwarding strategy for 5GS. By managing the time-triggered (TT) flow transmission time intervals on TSN translator and allocating enough frequency resources, the 5GS can imitate the CQF operation and seamlessly connect with nearby CQF enabled TSN bridges. Due to the complexity of resource scheduling across CQF-based TSN-5GS, traditional algorithms, such as dynamic programming and heuristics, are too time cost to search the optimal scheduling results. To improve the scheduling efficiency, we design a DRL-assisted time-frequency resource scheduling algorithm, namely DTF, to jointly allocate time interval and frequency resources for each

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TT flow. Different from the traditional DRL resource scheduling algorithms that separately perceive time intervals and link capacity utilization, DTF embeds with a two-dimensional matrix of link capacity utilization to efficiently learn the relationship between time intervals and capacity utilization.

Experimental results show that our proposed DTF algorithm in CQF-based TSN-5GS can increase the number of schedulable TT flows by 54.2% and 3.5x than DeepCQF and Naive, respectively. DTF can also achieve the best load balance with lower computational time cost. The contributions of this paper are listed as follows:

- We propose a time-aware packet forwarding strategy for 5GS to support the integration with CQF-based TSN. By managing the flow transmission time intervals on TSN translator and frequency resources, 5GS can seamlessly communicate with nearby CQF enabled TSN bridges.
- We design a DRL-assisted time-frequency resource scheduling algorithm, namely DTF, applied into CQF-based TSN-5GS for the first time. DTF embeds with a two-dimensional matrix of link capacity utilization to efficiently learn the relationship between different time intervals and capacity utilization.

The remainder of this paper is organized as follows. Section II introduces the architecture of TSN-5GS and our proposed strategy for 5GS integrated with CQF-based TSN. Section III formulates the joint time-frequency scheduling problem in TSN-5GS. Section IV designs the DTF algorithm for network resource scheduling. Section V evaluates the scheduling performance. Finally, our work is summarized in Section VI.

II. CQF-BASED SYSTEM MODEL FOR TSN-5GS

This section describes the architecture of TSN-5GS. To support integration with CQF-based TSN standard, we propose a time-aware packet forwarding strategy for 5GS.

A. TSN-5GS Architecture

The centralized architecture of TSN-5GS is defined by the 3rd Generation Partnership Project (3GPP) shown in Fig. 1. In data plane, the 5GS is regarded as a logical TSN bridge embedded into wired TSN system. 5GS is able to support most of the relevant TSN protocols, such as generic precision time protocol (gPTP), network configuration protocol (NETCONF), and VLAN-tagged stream filtering protocols. This is achieved by introducing the device-side and network-side TSN translators (i.e., DS-TT and NW-TT, respectively) in 5GS. The TSN translators can accurately compute the time difference between TSN and 5GS domains, and utilize the hold and forward buffering mechanism [7] to reduce the flow transmission latency and jitter (i.e., latency variance) from end to end. The real-time transmission guarantee across the entire system owes to network resource scheduling for each flow planned by the controller in control plane. Centralized network configuration (CNC) is the main component of controller. The end-to-end network resource scheduling task is computed by CNC equipped with efficient resource scheduling algorithms. All the TSN entities, including TSN hosts and bridges in the

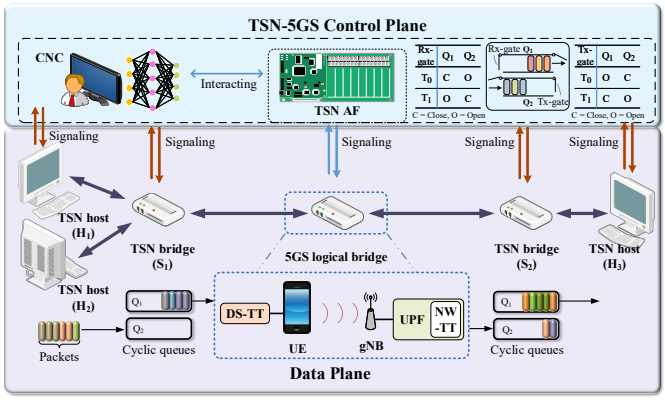

Fig. 1. Architecture of TSN-5GS supporting CQF mechanism.

management domain, provide exposure interfaces toward CNC to receive the signaling for configuration. The TSN application function (AF) communicates with CNC and delivers the port and bridge information to TSN translators in 5GS. Based on the architecture, we focus on the network resource scheduling problem with CQF-based TSN-5GS in this paper.

B. CQF-based System Model

CQF mechanism is an ease of use queue model in TSN to provide the deterministic transmission guarantee for TT flows across the network. We define the set of TT flows as $\mathcal{F}$. Each TT flow $f_i \in \mathcal{F}$ is featured by its flow period p_i, source host src_i, destination host dst_i, flow path pth_i, packet size s_i, packet number n_i, requirements of end-to-end latency $\mathcal{L}_i$, and jitter $\mathcal{J}_i$. Hence, the flow tuple is

$$f_i = \{p_i, src_i, dst_i, pth_i, s_i, n_i, \mathcal{L}_i, \mathcal{J}_i\}. \tag{1}$$

To shape the TT flows, CQF maintains two cyclic queues Q_1 and Q_2 periodically performing the queuing operations in each hyper cycle. As depicted by Fig. 1, we assume two time intervals T_0 and T_1 within a hyper cycle. During the even time interval T_0, the Q_1 opens the Tx-gate to send flows, and Q_2 opens the Rx-gate to receive the flows. During the odd time interval T_1, the queues operate conversely. IEEE 802.1 Qch standard [2] requires that all the sending flows from the upstream node in a time interval must be received by the downstream node within the same time interval. The received flows will continue to be forwarded in the next time interval. As such, the transmission delay between any two adjacent nodes can be constrained by one time interval. However, due to the different working mechanism between 5GS and wired TSN system, the CQF mechanism in TSN cannot be directly applied into TSN-5GS. Hence, we investigate the new approach to support the CQF mechanism from end to end in TSN-5GS.

Regarding the 5GS as a logical TSN bridge, we propose a time-aware packet forwarding strategy for 5GS to imitate the CQF operation in the time dimension to connect with the adjacent TSN bridges. Specifically, to ensure all the sending flows from TSN bridge being received by 5GS within one time interval T_ξ, enough frequency resources (i.e., sub-carriers of

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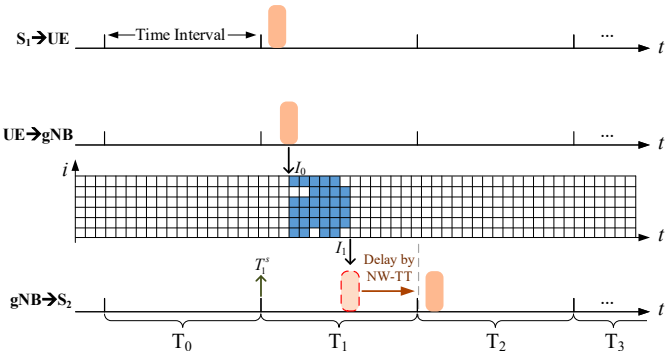

Fig. 2. An example of using time-aware packet forwarding strategy for 5GS.

orthogonal frequency division multiplexing (OFDM) system) should be allocated for transmitting the flows during T_ξ. Based on the hold and forward buffering mechanism of TSN translators, the received flows on 5GS device are stored into the buffer of TSN translator and wait to be sent until the next time interval. Fig. 2 describes an example of utilizing our proposed strategy. A packet of flow is sent from TSN bridge S_1 to UE during time interval T_1. After processing on UE, the packet is transmitted at timestamp I_0. Carried by a set of sub-carriers, the packet is received by 5G base station (gNB) at timestamp I_1 in the same time interval T_1. Then, the received packet waits on NW-TT, and continues to be forwarded at the next time interval T_2. In the process, the delays between DS-TT/NW-TT and UE/gNB are neglected because the link distance between them is short and bandwidth is large enough in our system. Based on our proposed strategy, TSN-5GS supports the CQF mechanism from end to end. Thus, the end-to-end latency $\mathcal{D}$ and jitter Δ only depend on the number of hops and time interval length, which is described by

$$\begin{cases} (\mathcal{N}-1)\times\mathcal{T} \le \mathcal{D} \le (\mathcal{N}+1)\times\mathcal{T}, \\ \Delta \le 2\times\mathcal{T}, \end{cases} \tag{2}$$

where $\mathcal{N}$ denotes the number of hops that the flow goes through, and $\mathcal{T}$ is the length of time interval.

III. Problem Statement and Formulation

This section describes the joint time-frequency resource scheduling problem and then makes the problem formulation.

A. Joint Time and Frequency Resource Scheduling Problem

Based on the CQF mechanism, the flow transmission through TSN-5GS faces both the time and frequency scheduling problems. First, the time interval scheduling problem is described by Fig. 3. In time interval T_0, the TSN host H_1 sends two packets, denoted as $pkt_{1.1}$ and $pkt_{1.2}$, respectively to TSN bridge S_1. And TSN host H_2 sends one packet $pkt_{2.1}$ to S_1. According to the network topology in Fig. 1, S_1 will forward the packets toward 5GS in the next time interval T_1. We assume that each time interval can accommodate two packets at most. As such, the packet $pkt_{2.1}$ will be dropped by the S_1 because it exceeds the time interval capacity of T_1. In contrast, if we postpone the sending time of $pkt_{2.1}$ to time

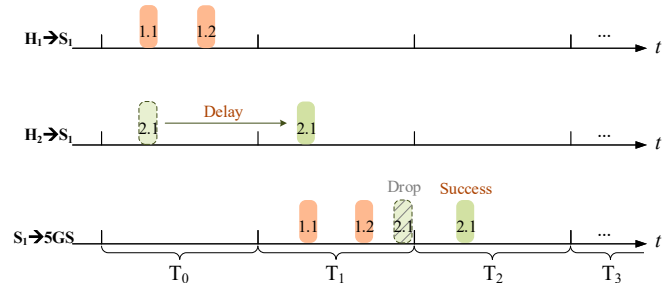

Fig. 3. Time interval scheduling for each flow.

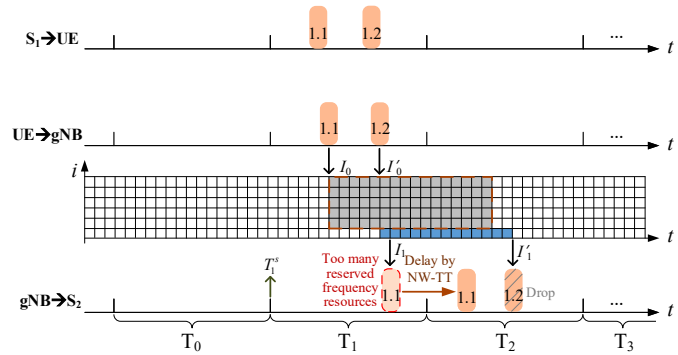

Fig. 4. 5GS bandwidth allocation for each flow.

interval T_1 on the host H_2, the $pkt_{2.1}$ will be successfully forwarded during time interval T_2 by S_1. Hence, the time interval scheduling from the TSN host for each flow is very important. Second, Fig. 4 describes the frequency resource scheduling problem. When forwarding packets $pkt_{1.1}$ and $pkt_{1.2}$ from UE to gNB, the sub-carriers in 5GS should be allocated to the packets. However, too many sub-carriers are reserved for carrying $pkt_{1.1}$, and $pkt_{1.2}$ cannot acquire enough 5GS bandwidth to be transmitted within the time interval T_1. This leads to packet loss. Therefore, both the time interval and frequency resources need to be scheduled to improve the network performance.

B. Problem Formulation

We formulate the joint time-frequency resource scheduling problem mentioned in Section III-A as the multiple objective optimization problem. The objective goals include maximizing the number of schedulable flows and system load balance. To measure the scheduling performance on the objective goals, we define functions $\phi(\cdot)$ and $\mathscr{C}(\cdot)$ that indicate the successfully scheduled flows and load balance, respectively. Hence, the objective goals can be formulated as

$$Maximize \quad \alpha\sum_{i=1}^{|\mathcal{F}|}\phi_i(\xi,\mathcal{B}) - \beta\mathscr{C}(f_i,\zeta), \tag{3}$$

where $|\mathcal{F}|$ is the number of flows, and $\alpha,\beta\in[0,1]$. If the flow f_i can be successfully scheduled by allocating the time interval T_ξ and 5GS bandwidth $\mathcal{B}$, then $\phi_i(\xi,\mathcal{B})=1$, otherwise zero.

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The function $\mathscr{C}(f_i, \zeta)$ computes the standard deviation of time interval utilization, and it has

$$\mathscr{C}(f_i, \zeta) = \sqrt{\frac{1}{\zeta}\sum_{\xi=0}^{\zeta-1}(\rho_\xi - \bar{\rho})}, \tag{4}$$

where ζ is the number of time intervals in one hyper cycle. ρ_ξ denotes resource utilization of time interval T_ξ, and $\bar{\rho}$ is the average time interval utilization. In addition, the optimization problem is subjected to several constraints. To ensure that the TSN-5GS periodically performs CQF operations, the hyper cycle $\mathcal{H}_c$ is defined as the least common multiple of all the flow periods. Let $LCM(\cdot)$ be the function that computes the least common multiple. Hence,

$$\mathcal{H}_c = LCM(\{p_1, p_2, p_3, \ldots, p_{|\mathcal{F}|}\}). \tag{5}$$

The granularity for resource scheduling with CQF mechanism is each time interval. It means that all the flow periods should be divisible by the predefined time interval. Thus, the maximal length of time interval $\mathcal{T}$ should be the greatest common divisor of all the flow periods. Let $GCD(\cdot)$ be the function of greatest common divisor. Hence, it has

$$\mathcal{T} \leq GCD(\{p_1, p_2, p_3, \ldots, p_{|\mathcal{F}|}\}). \tag{6}$$

To guarantee the high-reliable flow transmission across the network, the total size of packets sending at each time interval should be no larger than the maximal time interval capacity $\mathcal{T}_c$. Let the function $\psi_\xi(\cdot)$ denote whether the time interval T_ξ is allocated for the packet. It means $\psi_\xi(\cdot) = 1$ if the packet is mapped into T_ξ, otherwise zero. Therefore, this capacity constraint can be formulated as

$$\sum_{i=1}^{|\mathcal{F}|}\sum_{m=1}^{n_i} s_i \times \psi_\xi(pkt_i^m) \leq \mathcal{T}_c, \tag{7}$$

where the m-th packet of flow f_i is denoted as pkt_i^m. Specifically, the time interval capacity $\mathcal{T}_c$ depends on link bandwidth. In wired TSN network side, the $\mathcal{T}_c$ is computed by

$$\mathcal{T}_c = \eta \times \mathcal{W} \times \mathcal{T}, \tag{8}$$

where $\eta \in (0, 1]$ is the efficiency of time interval utilization, and $\mathcal{W}$ denotes the link bandwidth. In 5GS side, the $\mathcal{T}_c$ is related to 5GS bandwidth $\mathcal{B}$, time interval length, and signal to interference plus noise ratio γ. Hence, it has

$$\mathcal{T}_c = \eta \times \mathcal{B} \times \log_2(1 + \gamma) \times \mathcal{T}, \tag{9}$$

where γ relies on the power of transmitter ϑ, channel gain $g_{\ell\ell}$ between the UE and gNB, and interference from other links. As a result, the γ is described by

$$\gamma = \frac{\vartheta \times g_{\ell\ell}}{1 + \sum_{\ell' \in L} \vartheta' \times g_{\ell'\ell}}, \tag{10}$$

where ℓ' is the link that interferes ℓ, and $g_{\ell'\ell}$ is the gain of interference from ℓ' to ℓ. The transmission power on link ℓ' is denoted by ϑ'. As gNB schedules the flows in each transmission time interval (TTI), the transmission time for each packet between UE and gNB should be no later than a TTI length Γ, that is $I_1 - I_0 \leq \Gamma$. Moreover, based on our proposed time-aware packet forwarding strategy in 5GS, any packets sent from UE/gNB at time interval T_ξ must be received by gNB/UE within the same time interval. Let T_ξ^s be the start timestamp of T_ξ shown in Fig. 2. Hence, $I_1 - T_\xi^s \leq \mathcal{T}$. And the received packets at time interval T_ξ cannot be forwarded until the next time interval $T_{\xi+1}$ arrives, that is

$$\mathcal{T} \leq I_1 - T_\xi^s + \delta \leq 2 \times \mathcal{T}, \tag{11}$$

where δ is the time delay on TSN translators. For each TT flow, the sending time $f_i.\tilde{o}$ from the host should be no later than its flow period. This constraint can effectively avoid the flows in different periods preempting the limited resources at the same port. Thus it has $0 \leq f_i.\tilde{o} \leq p_i$.

To satisfy with the deterministic requirements of end-to-end latency $\mathcal{R}_d^{(i)}$ and jitter $\mathcal{R}_j^{(i)}$, the transmission of each flow should be constrained by $D_i \leq \mathcal{R}_d^{(i)}, \Delta_i \leq \mathcal{R}_j^{(i)}$.

IV. DTF: DRL-ASSISTED TIME-FREQUENCY RESOURCE SCHEDULING ALGORITHM

To solve the joint time-frequency resource scheduling problem, we transform the formulated problem into a Markov Decision Process (MDP). The core elements of MDP include state space, action space, and reward. The state space is composed by the flow features o_t^f and time-frequency resource utilization o_t^u at each step t. Therefore, the state $o_t = o_t^f \cup o_t^u$. Specifically, the flow features in o_t^f are the same with Eq. (1),

$$o_t^f = \{p_t, src_t, dst_t, pth_t, s_t, n_t, \mathcal{L}_t, \mathcal{J}_t\}. \tag{12}$$

The o_t^u reflects the utilization of time interval and frequency resources on each link. Considering the link capacity within each time interval can measure the time-frequency resource utilization, we define $u_{T_\xi}^{\ell_i}$ as the utilization of capacity on link ℓ_i during time interval T_ξ. Thus,

$$\begin{aligned} o_t^u = \{&u_{T_0}^{\ell_1}, u_{T_0}^{\ell_2}, u_{T_0}^{\ell_3}, \ldots, u_{T_0}^{\ell_L}, \\ &u_{T_1}^{\ell_1}, u_{T_1}^{\ell_2}, u_{T_1}^{\ell_3}, \ldots, u_{T_1}^{\ell_L}, \\ &\vdots \qquad \vdots \qquad \vdots \\ &u_{T_\mu}^{\ell_1}, u_{T_\mu}^{\ell_2}, u_{T_\mu}^{\ell_3}, \ldots, u_{T_\mu}^{\ell_L}\}. \end{aligned} \tag{13}$$

The action space comprises a set of time intervals in one hyper cycle and the 5GS bandwidth waiting to be allocated for each TT flow. At each step t, the action a_t is to allocate one time interval T_t and 5GS bandwidth for flow f_t. Instead of directly allocating a large number of 5GS sub-carriers, we reduce the action space of 5GS bandwidth allocation. We use $\mathcal{B}_t \in \{0, 1\}$ to denote whether the bandwidth is allocated for f_t, i.e., $\mathcal{B}_t = 1$ if f_t can acquire the 5GS bandwidth, otherwise zero. Once $\mathcal{B}_t = 1$ holds, we allocate the 5GS bandwidth with lower limit that supports the flow transmission in the valid duration. Hence, the a_t is described by $a_t = \{T_t, \mathcal{B}_t\}$.

The immediate reward r_{t+1} at each step t is related to the scheduled flow number and load balance. Let r_{t+1}^a and r_{t+1}^b be the shaping reward of scheduled flows and load balance, respectively. Thus, the total reward is $r_{t+1} = \sigma^a \times r_{t+1}^a +$

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$\sigma^b \times r^b_{t+1}$, where $\sigma^a, \sigma^b \in [-1, 0]$. In specific, r^a_{t+1} relies on the flow number, and r^b_{t+1} is the difference of load balance between two consecutive steps. Hence,

$$\begin{cases} r^a_{t+1} = \big|\Theta_{t+1} - |\mathcal{F}|\big| - \lambda^a \big|\Theta_t - |\mathcal{F}|\big|, \\ r^b_{t+1} = \big|max(o^u_{t+1}) - \Xi\big| - \lambda^b \big|max(o^u_t) - \Xi\big|, \\ r^a_{t+1} \leq 0, r^b_{t+1} \geq 0, \end{cases} \tag{14}$$

where $\lambda^a, \lambda^b \in [0, 1]$. The scheduled flow number at step t is denoted as Θ_t. The $max(o^u_t)$ represents the maximal usage of link capacity, and Ξ is the ideal occupancy of link capacity. Moreover, if all the flows can be successfully scheduled, an additional large reward can be obtained.

Based on the transformed MDP, we design a DRL-assisted time-frequency resource scheduling algorithm, namely DTF, to jointly schedule the time interval and frequency resources in TSN-5GS. In particular, a two-dimensional matrix of link capacity utilization according to Eq. (13) is embedded into DTF as the observation o_t. Each row of the matrix denotes a time interval, and each column indicates a link. The $u^{\ell_i}_{T_\xi}$ is the intersection point of each row and column. This matrix can effectively assist the agent to learn the relationship between different time intervals and capacity utilization. To extract the key features from the two-dimensional capacity utilization matrix, the convolutional neural network (CNN) can be introduced for training. The pseudo code of DTF is listed in Alg. 1 In line 1-8, the agent observes the current capacity utilization matrix o_t, and makes a time-frequency resource allocation decision a_t based on the mapping policy $\pi : O \rightarrow A$. As we use the value-based DRL framework, the action with maximal Q-value will be chosen. The Q-value, which is computed by the action-value function $Q^\pi(o, a)$, can reflect the long-term revenue, where

$$Q^\pi(o, a) = E\left[r_{t+1} + \gamma Q^\pi(o_{t+1}, a_{t+1}) \big| o_t = o, a_t = a, \pi\right]. \tag{15}$$

After taking the action, the immediate reward and next state can be obtained, and they are stored into replay buffer. In line 9-18, the agent samples a batch of trajectories from replay buffer and trains the neural networks. DTF maintains two CNNs (i.e., evaluation and target CNNs) with the same structure. The evaluation CNN is trained by updating parameter ω to minimize the loss, where the loss function is $l_o = \frac{1}{\mathcal{E}^s_b} \sum_{j=1}^{\mathcal{E}^s_b} (y_j - Q(o_j, a_j; \omega))^2$. In specific,

$$y_j = \begin{cases} r_j, \quad if \;\; done, \\ r_j + \gamma Q'(o'_j, \arg\max_{a'_j \in \mathcal{A}} Q(o'_j, a'_j; \omega); \omega^-), otherwise, \end{cases} \tag{16}$$

and the parameter ω^- of target CNN can be replaced by ω of the evaluation CNN every κ steps.

V. Simulation and Performance Evaluation

This section describes the simulation settings of CQF-based TSN-5GS and evaluates the scheduling performance of our proposed algorithm.

Algorithm 1: DTF algorithm for joint time-frequency resource scheduling.

Input: Empty replay buffer $\mathcal{E}_b$, Initial network parameters ω and ω^-, Replay buffer size $\mathcal{C}_r$, Training batch size $\mathcal{E}^s_b$, Action space $\mathcal{A}$.
Output: Optimal policy π^*.

1 **for** *episode* $e \in \{1, 2, \ldots, e_p\}$ **do**
2 Initialize the state o;
3 **for** *step* $t \in \{1, 2, \ldots, s_t\}$ **do**
4 Allocate time-frequency resources for flow f_t based on $Q(o_t, a_t; \omega)$ values;
5 Receive an immediate reward r_{t+1};
6 Transit to the next state o_{t+1};
7 Store tuple $(o_t, a_t, r_{t+1}, o_{t+1}, done)$ into $\mathcal{E}_b$;
8 Replace the oldest tuple if $|\mathcal{E}_b| > \mathcal{C}_r$;
9 Sample a minibatch of $\mathcal{E}^s_b$ tuples $(o, a, r, o', done)$ from $\mathcal{E}_b$;
10 Define $a^{max}(o'; \omega) = \arg\max_{a' \in A} Q(o', a'; \omega)$;
11 Determine $y_j = \begin{cases} r, & if \; done, \\ r + \gamma Q'(o', a^{max}(o'; \omega); \omega^-), & otherwise; \end{cases}$
12 Perform a gradient decent step with loss $L = E[(y_j - Q(o, a; \omega))^2]$;
13 Reset $\omega^- = \omega$ every κ steps;
14 **if** *done* **then**
15 break;
16 **end**
17 **end**
18 **end**

A. Settings for TSN-5GS and DTF algorithm

Settings for CQF-based TSN-5GS: The benchmark simulation topology, which has 3 TSN hosts, 2 TSN bridges, and one 5GS using frequency-division duplex (FDD) mode, is shown in Fig. 1. The link bandwidth of wired TSN is set to 1000Mbps. The frequency band of 5GS is 3.7∼3.8GHz, and 5GS bandwidth is 40MHz. The transmitting power of gNB and UE are 40dBm and 20dBm, respectively. In terms of the indoor hotspot LOS scenario [8], the channel gain is $PL(dB) = 32.4 + 17.3 \log_{10}(l_{5G}) + 20 \log_{10}(f_c)$, where we set $l_{5G} = 100m$ denoting the distance between UE and gNB, and f_c is center frequency. The Gaussian noise power is -104dBm. 1000 TT flows with periods {4ms, 8ms, 16ms} are scheduled in TSN-5GS. The packet size is in the range of 50B∼1KB [9]. The time interval is set to 4ms, and TTI in 5GS is 1ms.

Settings for DTF algorithm: We use two convolution layers, followed by one fully-connected layer in CNNs. Based on our experience, the kernel size of each layer is set to 3×3, and the stride is 1. The sizes of sample batch and replay buffer is set to 32 and 1e4, respectively. We adopt the Adam optimizer with 1×10^{-4} learning rate. The discounting factor γ is set to 0.9 to acquire the long-term reward. For the shaping reward, we set the parameters $\sigma^a = -1, \sigma^b = -0.1$

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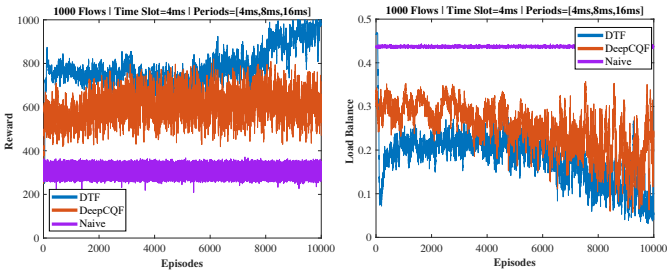
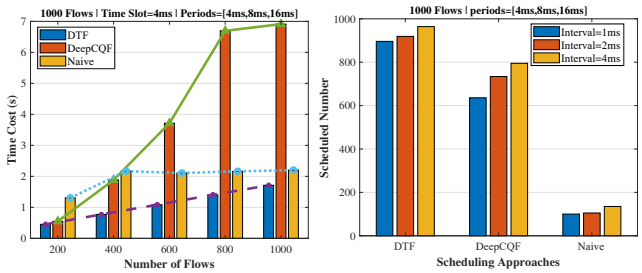


(a) (b)

Fig. 5. Training process. (a) Training on the total reward; and (b) Training for system load balance.



(a) (b)

Fig. 6. scheduling results. (a) Comparison of time cost; and (b) Comparison of schedulable flow number.

to emphasize the importance of schedulable flow number. And the other parameters λ^a and λ^b are both set to 1.

B. Performance Evaluation

We evaluate the scheduling performance of our DTF algorithm on Python implementations. The experimental results are described as follows.

Convergence comparison: The training process of different scheduling algorithms is depicted by Fig. 5. We compare our DTF algorithm with other benchmarks, which are DeepCQF [10] and Naive [11], respectively in CQF-based TSN-5GS. The DeepCQF schedules time interval resources by using two fully-connected layers, and each layer has 1000 neurons. The naive randomly allocates time and frequency resources for each TT flow. After 10,000 episodes, Fig. 5(a) shows that DTF improves 54.2% and 3.5x schedulable flow number than DeepCQF and Naive, respectively. This is because DTF can jointly consider time interval and frequency resources allocation for each flow, while DeepCQF only schedules time intervals and ignores the frequency resources that leads to the inadequate resource utilization. Moreover, the two-dimensional capacity utilization matrix embedded in DTF can effectively assist the agent to learn the relationship between time intervals and link capacity utilization. Fig. 5(b) shows the DTF also achieves the best load balance.

Scheduling Performance: As shown in Fig. 6(a), the computational time cost for DTF and Naive are almost the same as the number of TT flows increases. DeepCQF spends more time to compute the scheduling results because of the iterative searching process with exchanging mode. By changing the time interval length in Fig. 6(b), the DTF can always schedule the most number of flows. In addition, the schedulable flow number increases as the time interval length gets larger because more capacity can be used to accommodate TT flows in each time interval.

VI. Conclusion

In this paper, we have investigated the joint time-frequency resource scheduling problem in CQF-based TSN-5GS. We have proposed a time-aware packet forwarding strategy for 5GS to support CQF scheduling from end to end. Our strategy manages the network resources in finer granularity compared with the existing methods focusing on other queue models. To efficiently improve the scheduling performance, the DTF algorithm has been designed. Experimental results show that our proposed scheduling algorithm can achieve a good trade-off between runtime and schedulability. Different from traditional DRL resource scheduling algorithms, our DTF can better learn the relationship between time intervals and link capacity utilization. For the future work, we will investigate mixed-flow transmission in CQF-based TSN-5GS.

Acknowledgment

This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB2901302, in part by the National Natural Science Foundation of China under Grant 62201029, in part by the China Postdoctoral Science Foundation under Grants 2022M710007 and BX20220029.

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